

COURSE TITLE : DIGITAL PRINTING
COURSE CODE : 4102
COURSE CATEGORY : A
PERIODS/ WEEK : 5
PERIODS/ SEMESTER : 70
CREDIT : 5

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Basics of digital printing	20
2	Ink jet and laser technology	18
3	Modern application of digital printing	17
4	Digital offset printing	15
TOTAL		70

COURSE OUTCOMES:

G .O.	ON THE COMPLETION OF THE STUDY OF THIS MODULE STUDENT WILL BE ABLE :
1	To comprehend the modern digital printing technologies
2	Comprehend the working of inkjet and laser printing technology
3	To Understand the application of digital printing in modern industry
4	To understand the working principle of digital offset printing machines

MODULE I BASICS OF DIGITAL PRINTING

1.1.0 To comprehend the modern digital printing technologies

- 1.1.1. To state the concept of digital printing
- 1.1.2. To explain different digital printing technologies
- 1.1.3. To define impact and non impact printing
- 1.1.4. To list obsolete digital printers
- 1.1.5. To identify the use and working of Scanners
- 1.1.6. To explain characteristics of digital ink and tonner.

MODULE II INK JET AND LASER TECHNOLOGY

2.1.0 Comprehend the working of inkjet and laser printing technology

- 2.1.1 To define inkjet printing
- 2.1.2 To explain the working principles of ink jet printing
- 2.1.3 Drying process of inkjet printing ink
- 2.1.4 To list out various parts of ink jet printer
- 2.1.5 To define laser printing
- 2.1.6 To explain the working principle of laser printing

- 2.1.7 To compare inkjet and laser printers
- 2.1.8 To explain wide format digital printers

MODULE III MODERN APPLICATION OF DIGITAL PRINTING

3.1.0 To understand the application of digital printing in modern industry

- 3.1.1 To define use and advantage of digital printing in proofing
- 3.1.2 To identify different general application of digital printing
- 3.1.3 To explain use of digital printing in advertisements
- 3.1.4 To define specialized application of digital printing

MODULE IV DIGITAL OFFSET PRINTING

4.1.0 To understand the working principle of digital offset printing machines

- 4.1.1 To define digital offset printing
- 4.1.2 To identify functional parts of digital offset printing machine
- 4.1.3 To explain the working of digital offset printing machines
- 4.1.4 To identify the applications of digital offset machine
- 4.1.5 To define types of offset printing machines
- 4.1.6 To list out the manufactures of digital offset printing machines
- 4.1.7 To explain direct imaging presses

CONTENT DETAILS

MODULE I

Definition of digital printing. Basic digital printing work flow. Digital printing application. Advantages of digital printing – cost, consistency, storage, size, artistic control, freedom and flexibility. Digital image - Pixel and bit depth.

Digital printing technologies (introduction only) – electro photography, Ink jet, thermography, toner jet, magnetography. Impact and non impact digital printers. Dot matrix printers (brief introduction only).

Scanner – Types of scanners. Scanner resolution, meaning of scanner resolution. Relation between scanner resolution and output quality. Working of flatbed scanners. CCD arrays.

Characteristics of tonner powder. Characteristics of ink jet ink.

MODULE II

Definition of ink jet printing. Classification of ink jet printing -Drop on demand ink jet , continuous ink jet. Brief explanation of working principles of – Binary deflection, multi deflection, thermal ink jet, piezo ink jet, electrostatic ink jet, solid ink.

Cartridge - print head.

Definition of laser printing. Brief explanation of working principle of Laser printer.

Comparison of ink jet and laser printing, advantages and disadvantages. Names of major ink jet and laser printers manufacturers.

Definition of wide format digital printing. Application of wide format printing. Wide format printing machines. Names of major wide format digital printing machine manufacturers.

MODULE III

Use of digital printing as proofing method. Digital printers used for proofing – inkjet, laser. Advantages of using digital printing as proofer.

Digital printing substrates – paper, board, plastics, cloth, metal foils/sheets, glass, ceramic, etc.

Comparison between Digital printing and offset printing, advantages and disadvantages.

Use of digital printing for various purposes – general printing purpose, advertising, print on demand, personalization, variable data printing, etc.

Use of digital printing for advertising purpose – examples and their effects.

LED printer.

MODULE IV

Definition of digital offset printing. Functional parts of a digital offset printing. Working of digital offset printing machine. Application of digital offset printing. Name the major manufactures of digital offset printing machines.

Direct imaging technology. Direct imaging presses. Features of direct imaging presses. Plates in direct imaging press.

Reference :		
Author	Title	Publishers
Harald Jhonson	Mastering digital printing second Edition	
AGFA	Digital color Prepress	AGFA
Dr R K Molla	Electronic colour separation	R.K. Printing & Publishing company, USA
Adams J.M	Printing Technology 5 th edition	Delmar Publishers, NewYork
Helmut Kipphan	Hand book of print media	Springer Science & Business Media
Howard M. Fentan	On demand Digital Printing	GATF